Mark Scripsick

(903) 312-0084 | markscottscripsick@gmail.com | linkedin.com/in/mark-scripsick-0a7370392/ | markscripsick.github.io

Education

Texas A&M University Aug. 2025 – May 2029
B.S. Mechanical Engineering — GPA: 3.89 *College Station, TX*

Experience

Structures Team Member Aug. 2026 – Present
Texas A&M University Rocket Engine Design Team (RED) *College Station, TX*

- Assisting in the design, manufacturing, and validation of a 6061-T6 Al test stand rated for a gimballed vertical thrust load of 1,500 lbf with a FOS of 4 on yield, validating the design through Ansys transient and static simulations
- Fabricating shielding for avionics components to protect sensitive electronics during October hot-fire testing
- Collaborating with avionics, fluids, and thrust chamber sub-teams to ensure design integration across projects

Engineering Document Control May 2025 – August 2025
Baker Tank Company *Arp, Texas*

- Modernized filing system front to back, creating online repositories for drawings and their associated order details and specifications smoothing communication between engineers, drafters, shop foremen, and customers.
- Updated current software to allow job schedule, job number, order specifications, date ordered and fulfilled, as well as customer data to integrate with past jobs in one secure location reducing logistical load among all departments.

Teaching Assistant April 2024 – August 2024
Panola College *Carthage, Texas*

- Collaborated with Dean of Occupational Therapy at Panola College to create a comprehensive lesson plan for Occupational Therapy students to develop proficiency with 3D printers and 3D scanners to create tools for patients during field visits.
- Taught lectures covering 3D printer usage and maintenance, STL to G-code conversion using slicing software, material considerations for everyday conditions, and using patient feedback to iterate and improve designs.

Projects

Pedal Box Assembly | *Independent CAD/FEA Project* 2026

- Designed and structurally analyzed a machined aluminum pedal-actuator assembly to withstand a 3,000 N transverse load, following Formula SAE guidelines.
- Through iterative analysis decided on 6061-T6 aluminum for structural rigidity, weight, and ease of machining while maintaining a minimum factor of safety of 2.0 for all parts to ensure safety of potential users.
- Validated stress concentrations at all points using mesh convergence studies to rule out point failures and ensure minimal displacement at all operable loads.

TURTLE Robotics – Articulating Arm | *Mechanical Designer, Texas A&M* Jan. 2026 – May 2026

- Designed and fabricated a motorized articulating arm using 3D-printed gears, custom structural components, and DC motors, then integrated it into the team's competition robot through cross-disciplinary coordination with CAD, circuit, and software sub-teams.

FREE Works Design Challenge | *Lead Designer* October 2025

- Designed a seatbelt locking mechanism assembly in SolidWorks, consolidating individual components into a final product that earned second place in the competition.
- Tested initial and individual part designs to improve inertial and speed-based stopping mechanisms, determined manufacturing tolerances, and evaluated material considerations for strenuous loading conditions.

Physical Therapy Aid Development | *Project Lead and Developer* May 2025

- Worked with local physical therapists to design and produce cost-effective therapeutic aids using 3D printing, saving patients tens of thousands of dollars while expanding access to medical equipment for underinsured individuals.
- Interviewed patients to identify condition-specific challenges, then designed and distributed personalized assistive devices, significantly reducing equipment acquisition time by bypassing the traditional insurance approval process.

Skills

Technical: Competent in SolidWorks (CAD, assemblies, and simulations), Ansys, Prusa Slicer, Lightburn, and Python
Tools: 3D Printing, Laser Engraving/Cutting, and Metal Arc Welding